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CASE REPORT

Multifocal auricular keloids following piercing: Clinical presentation, contributing factors, and therapeutic considerations in a young adult

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ABSTRACT

Introduction: Keloids are fibroproliferative scars arising from aberrant wound healing, with auricular sites being particularly susceptible after piercings.
Objectives: To report the case of a young woman who developed multiple auricular keloids following piercings and to highlight the factors contributing to their progression.
Methods: A 19-year-old woman with six keloid lesions on both ears following multiple piercings was clinically evaluated. The characteristics of the lesions and associated cutaneous findings were assessed.
Results: The largest lesions were firm, sessile, and clinically stable, while newer formations showed mild inflammation. Crusted, desquamative areas on the earlobes suggested concomitant allergic contact dermatitis to nickel sulfate. Persistent use of piercings after the onset of early lesions likely contributed to further keloid development, particularly on the right antihelix.
Conclusions: This case highlights the role of mechanical and inflammatory triggers in keloid progression and underscores the importance of early recognition, removal of precipitating factors, and consideration of multimodal therapy to prevent recurrence.

Key words: auricular keloids, piercing-related scarring, multifocal lesions, contact dermatitis, keloid management, multimodal therapy

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Introduction

Cutaneous scars represent the outcome of the tissue repair process following significant trauma. Keloids form when the fibrous tissue of the reticular dermis proliferates abnormally and wound healing proceeds in an exaggerated manner[1]. It is known, in fact, that superficial lesions that do not reach the reticular dermis never evolve into keloidal scars[2]. Keloids are more frequent in individuals with dark skin, and their genesis is attributable to multiple etiopathogenetic factors, including an imbalance between collagen synthesis and degradation, excessive mechanical tension at the suture site, and an altered immunological response[3-4]. At present, it remains difficult to modify the systemic and genetic risk factors involved in the formation of keloids and hypertrophic scars[2].

This condition has a negative aesthetic impact and may therefore cause discomfort in patients, as well as varying degrees of functional impairment. In the event of rupture, a keloid tends to transform into an ulcer with a poor healing tendency and, in some cases, may even undergo malignant evolution[1]. In the specific case of the external ear, keloids arise most frequently following piercings, infections, surgical procedures, or trauma, and are associated with a high risk of malformations, complications, and local recurrences. Owing to its anatomical structure, the external ear is exposed and easily subjected to trauma, atmospheric agents, and inflammatory processes[5].

The therapeutic options currently employed range from intralesional and topical treatments to surgical procedures, radiotherapy, and laser therapy[6]. Intralesional corticosteroids remain one of the main management strategies, although several other injectable agents are now available, including bleomycin, 5-fluorouracil, and botulinum toxin type A[7]. Surgical excision is often integrated with one or more of these adjuvant pharmacological treatments, and there is growing interest in refining both keloid removal techniques and methods of tissue closure and suturing. However, typical keloids, characterized by an intense inflammatory component, are not generally manageable with surgery alone. In such circumstances, a combined approach - consisting of excision followed by radiotherapy and/or corticosteroid administration - proves more effective[2].

Regarding radiotherapy, both external beam radiotherapy and interstitial brachytherapy are employed, delivered with variable low- to high-dose regimens. In clinical practice, PDL laser, cryotherapy, and compression dressings are also used[7].

Case presentation

A 19-year-old Italian woman presented to the Dermatology and Venereology Unit of the R. Dulbecco University Hospital for moderate acne and multiple keloid lesions localized to the auricular pinnae, which had developed following the placement of piercings. The patient, who had Fitzpatrick skin phototype III, reported no family history of keloids, hypertrophic scars, or cutaneous neoplasms.

On physical examination, six keloid lesions were identified: four on the right auricle and two on the left. The latter involved both sides of the helix, resulting in compression of the intervening space (Figure 1). The largest lesions, present on both helices, appeared as firm, sessile, non-infiltrated lesions with a fibrous consistency, and were painless upon palpation. The three largest lesions were clinically stable and showed no signs of inflammation, whereas the more recent lesions appeared mildly erythematous and inflamed (Figure 2). No locoregional lymphadenopathy was detected.

At the earlobes, where earrings were inserted, crusted and desquamative areas were observed, a finding compatible with possible allergic contact dermatitis, most likely related to sensitization to nickel sulfate (Figure 3). The patient's history revealed that the patient had not removed her piercings despite the appearance of the initial lesions, a behavior that presumably contributed to the development of additional keloids - particularly in the right antihelix, where the partially inflamed lesions described above were located (Figure 2).

Discussion

The molecular mechanisms underlying keloid formation are remarkably complex and cannot be reduced to a simple model of fibroblast hyperproliferation;

Figure 1. Large, clinically stable keloid lesions involving both sides of the left helix, causing compression of the intervening space.

Figure 2. Multiple keloids of the right auricle, including recently developed lesions on the antihelix showing mild erythema and inflammation.

rather, they reflect a dysregulated biological system in which genetic predisposition, aberrant mechanotransduction, chronic inflammation, and pathological cellular senescence converge and reinforce one another. Recent evidence demonstrates that keloid tissue harbours an expanded population of mesenchymal and pro-inflammatory fibroblast subsets exhibiting increased expression of senescence markers such as CDKN2A/p16 and, notably, enhanced phosphorylation of p53 at serine 15. Importantly, unlike physiological senescence, these cells do not undergo apoptosis but remain metabolically active and secrete a profibrotic senescence-associated secretory phenotype (SASP), thereby perpetuating a chronic inflammatory and fibrogenic microenvironment. Mechanistically, this phenomenon is linked to the nuclear stabilization of phosphorylated p53 through its interaction with FOXO4, which prevents p53 nuclear exclusion and impairs activation of the apoptotic cascade, ultimately allowing the persistence of senescent yet pathogenic fibroblasts[8].

Figure 3. Mature keloid of the helix associated with crusted and desquamative changes of the earlobe, suggestive of allergic contact dermatitis related to piercing jewellery.

Concurrently, integrated multi-omics analysis supported by Mendelian randomization has identified IGF1 as a causal risk factor for keloid development, demonstrating that activation of the IGF1–IGF1R axis stimulates the PI3K/AKT and MAPK pathways and functionally cooperates with TGF-β signalling to enhance collagen synthesis and cellular survival. In particular, the interaction between IGF1 and the integrin α6β4 complex promotes FAK activation and downstream engagement of mechanosensitive mediators such as YAP/TAZ, thereby establishing a self-reinforcing feedback loop between extracellular matrix stiffness and sustained fibroblast activation[9]. This mechanobiological circuit operates within a molecular landscape already dominated by hyperactivation of canonical and noncanonical TGF-β/Smad signalling and its crosstalk with Wnt/β-catenin and integrin-FAK-YAP pathways, collectively driving myofibroblast differentiation, excessive deposition of type I and III collagen, and impaired matrix degradation. Furthermore, epithelial-to-mesenchymal transition (EMT) and endothelial-to-mesenchymal transition (EndMT) contribute additional profibrotic cellular sources, while an immune microenvironment skewed toward M2 macrophage and Th2 polarization - enriched in IL-4, IL-13, and TGF-β1 - consolidates chronic fibrotic signaling[10].

Taken together, these findings support a conceptual framework in which keloids represent a pathologically stabilized tissue state sustained by dynamic interactions among genetic susceptibility (notably IGF1 signaling), aberrant mechanotransduction, persistent profibrotic signaling, and apoptosis-resistant senescent fibroblast niches. This integrated model suggests that truly effective therapeutic strategies should extend beyond mere suppression of collagen production and instead aim to concurrently disrupt pathological senescence circuits and mechanosensitive feedback loops that maintain the fibrotic phenotype.

Conclusion

This case underscores a rare and clinically significant presentation of multifocal, bilateral auricular keloids in a young adult without a familial predisposition or systemic risk factors. The simultaneous occurrence of multiple keloid lesions on anatomically distinct regions of both auricles, particularly following repetitive trauma from piercings and potential contact sensitization to nickel sulfate, illustrates the multifactorial nature of keloid pathogenesis.

The findings highlight the critical role of early intervention and the prompt removal of inciting mechanical and inflammatory stimuli to prevent lesion progression. Furthermore, the case reinforces the notion that auricular keloids - especially in multifocal and recurrent forms—pose a substantial therapeutic challenge, often necessitating a multimodal treatment approach combining surgical excision, intralesional pharmacotherapy, and, when indicated, adjuvant radiotherapy or laser-based modalities.

Given the aesthetic, functional, and psychosocial impact of auricular keloids, this case advocates for the development of standardized, evidence-based therapeutic algorithms and supports further clinical research into predictive markers and individualized treatment strategies to optimize long-term outcomes and reduce recurrence rates in this particularly vulnerable anatomical region.

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